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Chapter 25: The Discrete Leviathan—Definition, Historical Comparison, and the Structure of Power

Abstract

Digital services do more than connect existing needs with existing supply: they can also change what materials people encounter, what rewards they expect, and when they must make a choice. This chapter defines the Discrete Leviathan as a cross-layer network of key nodes: distinct actors, linked through technology, property rights, and contracts, continuously organize access, the presentation of information, and the feedback of outcomes, while those who depend on them can hardly exit these arrangements simply by switching to another service. The analysis moves from relational structure into human time and belief: who can set the problems, who controls the paths leading to outcomes, who can modify the rules, and by what routes the consequences of action return. The historical materials offer several distinct mechanisms. The medieval faculty of arts and printed textbooks organized the transmission of knowledge; the Ramist rearrangement of method changed the division of labor among invention, judgment, and expression; Bacon distinguished rapid assent from the gradual formation of reliable interpretation; Pascal placed the maintenance of belief in habit and repeated practice; and Bernoulli studied the weighing of evidence and the standards of action under incomplete certainty. The belief in Purgatory organized obligations that span death; commercial rule distributed gains and futures across different groups; and advertising, installment buying, and model changes together altered the meaning of buying and of waiting. These are not successive upgrades of one and the same mechanism, but historical clues for

understanding the contemporary assemblage. Common knowledge analysis shows that belief also depends on whether people can mutually confirm what they know; fiber analysis distinguishes changes in preference from changes in the expectations of future reward and changes in the conditions of action, so that identical short-run choices are not all read as a loss of patience. The distinctive power of this cross-layer network lies in its capacity to intervene in these conditions simultaneously, and to keep adjusting the environment with the behavior thereby produced. Generative artificial intelligence (AI) extends the scope of production and execution in this assemblage. Judgments about this network must therefore rest not only on business growth or single outputs, but also on how human purposes, intertemporal trade-offs, and the conditions of collective correction change in use.

Keywords

discrete Leviathan; relational topology; time preference; belief formation; cross-layer control; private governance; history of advertising; common knowledge; fiber; cognitive debt

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§1 From Using Services to Depending on Rules

§1.1 Why an Object beyond the Individual Platform Needs to Be Identified

A publishing house can own its content, yet it cannot decide on its own whether that content can be found through search; a merchant who operates through a platform can own its goods, yet it must approach customers through the entrances the platform prescribes; an app developer can write a program, yet it must still obtain the access conditions required for distribution, payment, and operation. These actors are not without the capacity to act. They can produce, invest, experiment, and compete—but whether their content reaches readers, whether their goods can be sold, and whether their programs are distributed and run depends to a considerable extent on rules set by other organizations.

What needs to be studied here is not merely how many users a given company has, but how control is transmitted along chains of activity. Content ownership, eligibility for display, recommendation placement, transaction gateways, and performance reports can belong to different actors; a single company can also control several of them at once. The former situation does not automatically produce open competition, and the latter does not automatically prove that all participants have lost their autonomy. What matters is this: who can change the rules, how such changes are transmitted to other links, and whether those who depend on them can refuse without losing the basic conditions for continuing their activity.

DeNardis's research on internet governance offers a starting point. The internet is physically widely distributed, and its governance bodies are more than one, yet concentrated control points still exist; some of these points are physically dispersed while being functionally quite centralized. Political and commercial conflicts over content can therefore be handled through infrastructure, technical standards, and intermediary arrangements (DeNardis, 2014). A distributed network and concentrated control can coexist; there is no need to presuppose that the internet was once entirely free of power in order to explain the concentration that came later.

This chapter calls one more specific configuration the **Discrete Leviathan**: a cross-layer network of key nodes formed by different organizations through technological dependency, property rights, and contracts, which—

in a number of important knowledge, communication, or transaction activities—continuously stipulates the eligibility for participation, shapes what information is visible, and uses the results of activity to adjust the conditions of participation in the next round, while the individuals and organizations positioned downstream in the network face markedly asymmetric costs of substitution and exit. "Cross-layer" refers to the control linkages among infrastructure, content production, information distribution, transaction, and execution.

This definition concerns a networked form of governance, not a ranking of company size. A firm can be very large yet lack control over the key conditions by which others access information and participate in activities; a node can operate only a single link yet acquire significant influence because other actors can hardly replace it. The standing of a node must be judged with respect to specific activities; it cannot be settled directly by market capitalization, listing status, or the company's name.

§1.2 What "Leviathan" and "Discrete" Each Denote

In Chapter XVI of *Leviathan*, Hobbes discusses how a multitude of people constitute one political person through relations of representation. He stresses that unity is formed in the representer; the representer may be one person, or an assembly of many persons that must take the opinion of the majority as the opinion of the whole (Hobbes, 1651/2024). This chapter borrows "Leviathan" in order to place relations that seem to be mere buying, browsing, and using back into the question of rule-making and obedience. The borrowing implies neither that platforms are modern states nor that using a service amounts to completed political authorization.

The first difference between the contemporary node network and Hobbesian unified representation is that no single subject exists that can act on behalf of all nodes, nor does any subject bear complete responsibility for the operation of the whole network. Cloud providers, model developers, device providers, and distribution platforms can cooperate and can also compete. Without sharing a common will, they can, through the access rules each of them sets, compose the set of conditions that dependents must face. The absence of a commander-in-chief does not mean the absence of relations of domination; the existence of domination does not mean that all nodes are executing one and the same plan.

"Discrete" carries two related but non-interchangeable meanings in this chapter. The first is that control and responsibility are distributed across different nodes: one visible outcome may have passed through several organizations, while those affected must separately understand contracts, technical rules, and complaint procedures. The second is that activity is partitioned into units that can be recorded and processed: a single impression, a single click, a rating, an account status become the basis on which business systems judge and allocate resources. People's lives do not thereby become sequences of independent events; the system primarily records and evaluates activity through these processable units. The former describes the distribution of governing subjects and responsibilities, and it is the principal basis for the name of this chapter; the latter describes how systems process activity, and it is a common operating feature of this form of governance—that activity is discretely recorded does not by itself prove that cross-layer domination has taken shape.

Personalized information distribution may in turn make it difficult for different participants to confirm what the others have seen. This is an epistemic consequence of discrete arrangements, not an established fact built into the definition. Shared information and public discussion can still exist; whether they have been weakened must be observed for specific platforms, activities, and groups. One cannot infer, from the fact that digital computation uses discrete representation, that public life is necessarily shattered; nor can one infer, from the

fact that information feeds differ, that society no longer has common knowledge.

This networked form of governance can also be called the **loss function Leviathan**. Named after the loss function, it may be formally defined as a networked form of governance composed of interdependent key nodes that need not share the same objectives: it translates human activity into recordable, classifiable, evaluable, and optimizable objects, and then, through centralized control over optimization targets, scales of evaluation, and the channels of their enforcement, continuously organizes the allocation of information, opportunity, resources, and action, so that the objectives of local organizations become the environmental conditions to which other actors must adapt; this capacity to shape, in turn, is continuously reproduced by behavioral feedback, infrastructural dependence, and the reinvestment of capital. This alternative name marks its operating mechanism: organizations convert part of their business objectives into optimization metrics, and then, through access, ranking, pricing, and enforcement rules, make those metrics the conditions of action to which other actors must adapt. What content gains visibility, what actions are rewarded, and which consequences enter the statistics are thereby shaped by optimization targets, proxy metrics, and weights—yet the loss functions used by models, the assessment metrics adopted by products, and the business objectives of organizations are not identical. In engineering terms these links must not be conflated with one another either: the reward is not the whole content of the loss function, a parameter update is not a revision of the objective, a sampling setting is not the loss function, and contracts and release rules are not neural-network weights; yet together they help determine which results are selected, amplified, discarded, and enforced. A system may improve on its own scale of evaluation while leaving consequences outside that scale to users or to society—technical loss and social loss do not necessarily move in opposite directions. It requires no unified will, nor does it require all nodes to use the same loss function: individual nodes can pursue different and even conflicting objectives; what must be identified is how these objectives acquire external force through facilities, access, and distribution channels. The people affected by these rules, and who bear their consequences, do not thereby acquire commensurate capacities for setting objectives, verifying, contesting, and correcting—this asymmetry is the problem that the governance analysis of this chapter and the algorithmic central bank proposal of Chapter 35 jointly confront.

§1.3 The Four Discriminant Dimensions and Their Different Standing

Identifying the Discrete Leviathan requires distinguishing the control conditions of a node from the maintenance conditions of the network. Serial control and substitution cost asymmetry serve, in the first instance, to judge whether a node possesses substantive dominance; trans-sovereign reach and capital structure reflexivity further explain how such dominance extends and maintains itself. The latter two must not be written as absolute properties that every node must satisfy. These four dimensions together constitute the discriminant function of this chapter; references in later chapters to the network of key nodes hit by the discriminant function of Chapter 25 §1 denote the network identified by applying these four tests item by item. The discriminant function is not a mathematically proven necessary-and-sufficient classifier, nor does it replace this chapter's criteria of governance mechanism—a hit identifies the object but does not by itself yield conclusions about the strength of control or its consequences.

Table 1. The four discriminant dimensions of the cross-layer network of key nodes

Dimension	The relations to be ascertained	What cannot be automatically concluded from it
Serial control	Which key links an activity passes through; who can change the conditions of access, presentation, pricing, or enforcement; whether the change is transmitted to other layers	That an activity must technically pass through a certain kind of facility does not mean it must pass through a particular firm
Substitution cost asymmetry	What it costs a dependent to migrate data, relationships, programs, and business; what it costs the node to lose this dependent	The existence of switching costs does not mean that no one can exit; feasible alternatives must be compared
Trans-sovereign reach	In which jurisdictions services, assets, contracts, data, and the people affected are distributed; which powers can reach which links	Operating across borders does not mean being above the law; that no single state can handle the whole chain does not mean it cannot intervene effectively
Capital structure reflexivity	How expected revenue supports financing and investment; how investment in infrastructure and gateways changes the position of control; how that position in turn shapes future revenue	Large outlays or a rising valuation do not, taken alone, prove a narrative bubble, irreversible concentration, or cognitive harm

For an actor to be identified as a key node, it must at least be shown to impose conditions with downstream effect along an important chain of activity, conditions that those affected can hardly circumvent at comparable cost. A judgment at the network level—that a Discrete Leviathan is present—must further show how multiple links jointly control information and participation; not every supplier with market power can be brought in.

For example, chips, cloud computing, models, and application gateways constitute an AI service chain that can be traced layer by layer, but this is not a list that has already proven every firm on it irreplaceable. The equipment and migration costs required for training a model and for running an everyday application are not the same. Concentration in one link becomes evidence of cross-layer control only when it is transmitted through actual supply relations, software dependencies, contractual restrictions, or gateway arrangements.

These dimensions also require distinguishing different forms of power. Property rights enable an actor to modify product rules directly; contracts can restrict the choice of counterparties; technological dependency can make replacing a solution costly. These relations may reinforce one another, but the analysis must distinguish their type and strength rather than read every connection as an equivalent degree of control. A capital relation is not automatically a relation of operational command either, and cooperation does not cancel the conflicts of interest among nodes.

§1.4 A Concrete Cross-Section: How Search and Advertising Are Connected

Caliskan, MacKenzie, and McGowan's study of Google's advertising ecosystem presents search as a production process of crawling, indexing, ranking, the delivery of results, and data-center support. What users query is not the unorganized web as a whole but the index this process provides and its ranked results. A site can refuse to be crawled, but for sites that depend on search to reach readers, such refusal simultaneously means losing an important source of visibility (Caliskan, MacKenzie & McGowan, 2026).

The same study points out that upstream components such as crawling and indexing are not traded as market services that can be purchased item by item, while downstream advertising slots are sold through a highly organized trading system. What competitors face is therefore not only competition in bids within a given auction, but also the costs of building the corresponding content, data, and infrastructure. Google, moreover, simultaneously acts in its advertising system as market organizer, intermediary, and service provider. These concrete connections explain how cross-layer control takes shape far better than the phrase "the platform is big."

There are real service gains here. The index lowers the cost of finding material; the advertising system connects dispersed buyers and sellers. But service gains and rule power are not mutually exclusive explanations. Precisely because these services are useful, other actors may organize their own activity upon them; thereafter, changing the conditions of the service can affect the business and life arrangements that have formed on top of it. This case shows that service gains can coexist with rule control across links. Whether it further constitutes a Discrete Leviathan still depends on how these controls connect different organizations, and on what feasible alternative paths the dependents have.

§1.5 How Topological Relations Enter Human Time and Belief

Borrowing the relational perspective of topology, this chapter first examines who is connected to whom, through which intermediaries one must pass, which channels of behavioral feedback, outcome verification, and appeal are open, and which channels are controlled by the same actor. Nodes can be individuals or organizations; connections can transmit information, allocate resources, confer qualifications, or enforce obligations. People's profiles and scores are representations this network uses; they cannot be taken directly as substitutes for relations among people. Drawing all of them as identical points and lines would conceal the very differences that need to be explained.

Mere connectivity does not yet entail any psychological consequence. For a merchant approaching customers through a platform, rules that remain stable over time and rules that can change at any moment do not mean the same thing for sustained investment; receiving a recommendation while being able to question its author and change the conclusion, and receiving one while only being able to accept the result or leave, are not the same relation either. The analysis must therefore specify the direction of influence, who has authority to decide, who must wait, and who bears the consequences. Topological analysis provides the relational skeleton; concrete institutions and temporal arrangements are what make it a structure capable of explaining action.

Such a structure affects action through at least two paths. The first changes the feasible choices: which materials are easy to obtain, which actions are rewarded, and what opportunities waiting forfeits. Even if people's purposes remain unchanged, their choices may change with them. The second operates within repeated participation: the explanations one is continually exposed to, the practices that are rewarded, and others' reactions gradually enter one's understanding of what is credible, worthwhile, and normal. At this point the environment

no longer only affects the realization of existing preferences; it may also take part in the formation of preferences and beliefs.

The two paths can reinforce each other and can also come apart. That a creator shortens the production cycle because income is about to be interrupted does not mean that he has come to prefer short-term projects; only by continuing to observe his standards of evaluation and his intertemporal choices can one judge whether this adaptation has further changed his preferences. The study of the Discrete Leviathan thereby goes beyond a static description of corporate power: what must be traced is how relations of control organize human experience, and how, with that experience, they continue to sustain themselves.

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§2 How Knowledge Is Organized: From University Training to the Sustained Formation of Belief

The historical materials that follow are used to compare mechanisms, not to construct a continuous lineage leading to digital platforms. Theoretical claims, organizational practices, and actual effects in these texts will be treated separately.

§2.1 The Medieval Faculty of Arts: The Institutional Production of Forms of Knowledge

In studying Ramus, Ong first returns the question to the medieval university. What he attends to is not the systems of a few theologians but the teaching tradition of the faculty of arts, centered on Paris and running from the thirteenth century to the early sixteenth. The faculty of arts organized training in grammar, rhetoric, logic, and natural philosophy. Teachers transmitted knowledge through courses and textbooks, while degree systems certified those who had received the training. Students later carried these forms of knowledge into teaching and other professions. Forms of knowledge therefore existed not only in the works of individual thinkers, but also in the institutions that continuously cultivated the next cohort of users (Ong, 1958/1983).

The classification of the trivium and the quadrivium cannot be taken directly as the actual curriculum here. Ong points out that the trivium—grammar, rhetoric, and logic—and the quadrivium—arithmetic, geometry, astronomy, and music—long remained an ideal frame of reference; in actual teaching, much of the study that followed logic belonged to natural philosophy and was not arranged neatly according to the seven arts. The distance between the ideal classification and teaching practice is itself part of the history of knowledge.

Ong stresses above all that, so that each new generation of students could learn, the universities continuously collated, simplified, and codified a complex scholarly tradition into curricula and textbooks. This organization brought evident gains: knowledge could be transmitted beyond the ties of individual teachers and students, and newcomers could learn along relatively stable paths. But convenience of teaching and adequacy for research are not the same standard. A system easy to teach does not necessarily preserve sufficient room for questioning and correction for those who learn later.

It follows that the relation between knowledge and people is not merely that authors write conclusions and readers choose to believe. The curriculum fixes the order of learning; the textbook fixes the shape of questions; the qualification fixes which performances gain recognition. An analysis of this relation need not first pronounce education pathological; what it must explain is how certain ways of thinking became, through organized training,

the default conditions of later judgment.

§2.2 The Ramist Rearrangement: How Invention, Judgment, and Expression Were Separated

In the sixteenth century, Ramus attempted to reorganize logic and rhetoric. In Ong's reconstruction, he concentrated the invention, judgment, and arrangement of arguments in dialectic or logic, restricting rhetoric chiefly to verbal ornament and the delivery of speech. The important change is not that a few fewer figures of speech were taught, but that knowledge formation and speech among people were assigned different standings: thought seems to be completable first within a method, and only then handed over to expression for transmission (Ong, 1958/1983).

Ong describes the Ramist arts of discourse as tending toward monologue. What he criticizes is not that question and answer were wholly absent from the classroom, but that the activity of discovering problems, entertaining objections, and forming judgment in the give-and-take among people lost its former theoretical place. In Ong's account, judgment increasingly came to be understood in terms of classification and arrangement; topics were placed into classifications and diagrams, and the order in the textbook was invested with epistemic authority.

Such a method still had to handle audiences unwilling to accept conclusions in the prescribed order. Ong records that Ramus distinguished clear teaching from the use of concealment, deferral, inversion, and similar devices to work upon an audience; in the latter use, the audience's resistance chiefly becomes an obstacle to be overcome. The earlier tradition of probable argument had allowed one to proceed from premises that were never fully certain to conclusions that remained disputable; Ramus's demand for certainty left him little room for this activity.

Yet in Ong's analysis, Ramist logic itself never achieved the rigor it claimed. It still borrowed rhetorical topics, and its actual conclusions could often be no more than probable; the neatness of diagrams and method could not by itself fill the gap in proof. This locates the question at a precise point: does method help participants judge further, or does an order convenient to present acquire the authority to terminate judgment?

The printed textbook reinforced this organizing capacity. Fixed pages, classifications, and charts could be copied repeatedly, and readers could use the knowledge apart from the occasion of the lecture. Ong treats printing as an important condition of this change, not as its sole cause. The same technology also preserved dissent, old editions, and arguments, so printing itself cannot be pronounced the end of dialogue. What must be traced is how the fixed text combined with teaching qualifications, the authority to interpret, and channels of response.

§2.3 Bacon: The Time Lag between Easy Assent and Worthy Belief

In the *Novum Organum*, Bacon distinguishes two paths by which general judgments are formed. One leaps swiftly from a few familiar instances to universal principles, and then uses those principles to explain everything else; the other forms generalizations gradually from dispersed material. He criticizes the first for anticipating nature, while acknowledging that it wins assent more easily: familiar instances gratify the imagination at once, whereas explanations built from a broad range of material often seem difficult and uncongenial to established habit (Bacon, 1620/1902).

This distinction cannot simply be renamed a quarrel between logic and rhetoric. Premature generalization can equally possess logical form; the fault may lie in starting points that have not been adequately examined. Bacon also notes that when a new instance runs counter to a principle, people preserve the principle with hasty

distinctions rather than revise it. Incorporating new material does not necessarily revise an earlier judgment; additional distinctions may instead serve to preserve the original principle.

Here a temporal problem different from the speed of transmission appears. Winning assent can be quick; assembling grounds sufficient to support judgment takes longer. If an organization allocates opportunity according to how much assent is won right now, the latter may lose its support before it is complete. This step is an institutional question this chapter raises on the basis of Bacon's distinction; it is not a reading of his epistemology as the direct cause of modern advertising.

At the same time, taking longer does not guarantee that a conclusion is reliable, and quick understanding does not mean being misled. What matters is whether that time is spent looking for counterexamples, checking conditions, and revising judgment—not speed itself. There is no conversion rate between speed of acceptance and quality of knowledge that can be applied apart from the object at hand.

§2.4 Pascal: Belief Is Not Formed in a Single Argument

In *The Art of Persuading*, written in the seventeenth century, Pascal distinguishes between accepting opinions through the understanding and accepting them through the will. He gives rules for defining terms, conceding premises, and unfolding proofs, while noting that what pleases people varies with person, circumstance, and time, so that the art of pleasing can hardly be mastered as stably as geometrical demonstration. He also sets divine truth apart from the human truths that these rules of persuasion address (Pascal, 2017).

The *Pensées* push the question further into how belief is sustained. Pascal discusses belief as sustained by understanding as well as by habit: a person cannot always keep all the proofs present before his eyes; judgments once understood need to be maintained in the practice of living. He describes habit as a power that carries the mind along with it, and notes that this power appears alike in religious life and in trades and everyday activities (Pascal, 1995, fragment 821).

Fragment 418, the Wager, writes this tension in the form of a dialogue. The interlocutor accepts the reasons for betting yet still cannot believe. Pascal thereupon turns to the practice connected with believers, asking him to learn the ways of those people and to weaken the passions obstructing faith through repeated action. This is not a proof of God's existence from expected gain; it is a discussion of how to act, and of how action takes part in the formation of belief, on the premise that existence has not been rationally proven.

Seen from the relational structure, what this arrangement changes is not merely a sentence entering the mind, but whom a person is continually exposed to, what activities are repeated, and how the consequences of action are experienced. Previously accepted reasons support practice, while repeated practice helps sustain belief. Such return can help a person hold to a considered choice, and it can also firm up judgments that have never been checked; habit by itself supplies no guarantee of truth for belief.

Nor does this process necessarily make people attend only to the present. Pascal's practice of faith could precisely require a person to rearrange present life for the sake of a more distant goal. His criticism of our constantly fleeing the present and lodging happiness in the future shows, in turn, that facing the future is not automatically living better (Pascal, 1995, fragment 47). What the historical clue indicates is not that patience necessarily decays, but that a person's temporal orientation can be reorganized by the relations of practice and belief in which he stands.

§2.5 Bernoulli: Judgment and Action under Incomplete Certainty

In Part IV of the *Ars Conjectandi*, Bernoulli places the study of probability within civil, moral, and economic affairs. He discusses the number and probative force of arguments, distinguishing arguments drawn from relations of cause, effect, and sign from arguments resting on human authority and testimony. His question is how to form the more prudent judgment when complete certainty is unavailable—not how to make an audience more readily accept a predetermined conclusion (Bernoulli, 1713/1968).

The seventh rule of Chapter II of Part IV requires that the wisdom of an action not be assessed by its outcome alone: imprudent actions can succeed by luck, and prudent actions can meet with bad results. The eighth requires that the more probable judgment not be treated as absolutely certain, whether toward oneself or toward others. Inferring the quality of judgment from outcomes, and inferring certainty from greater probability, are thereby already explicitly constrained in this early work on probability.

Immediately afterward, the ninth rule adds another provision: complete certainty is rarely attainable, and practice therefore needs to treat judgments that have reached moral certainty as if they were absolutely certain; it further proposes that it would be useful for definite limits of moral certainty to be set by public authority. Moral certainty here is a degree of conviction sufficient to support a practical decision, not a grade of the goodness or badness of conduct. The tension between it and the eighth rule is the tension between knowledge that remains incomplete and action that must continue.

Beyond proof and the art of pleasing, then, one must also weigh incomplete evidence, judge whether it already suffices for action, and keep room for revision. Different organizations can demand different degrees of evidence, allow different waits, and vest in different persons the right to decide when deliberation ends. Probability does not abolish judgment; it brings the question—on how much evidence, and bearing what consequences, may one act—more explicitly into institutions.

§2.6 From Judgments about the Future to Action in the Present

In Chapter VI of *Leviathan*, Hobbes describes deliberation as a succession of desire, aversion, hope, and fear around the consequences of action. He further notes that every human action is the beginning of a long chain of consequences, of which scarcely anyone's foresight reaches the end (Hobbes, 1651/2024). Those who see the consequences further and more reliably, through experience and reason, are also better able to judge and to advise. This discussion places action within continually forming expectations of consequences, rather than treating people as holders of fixed desires awaiting external stimuli.

Three questions can thus be separated: what a person believes will happen, what outcomes a person considers worth pursuing, and what a person can do now. Supplying new evidence may change the first; repeated practice and social evaluation may take part in the second; qualifications, resources, and rights bear directly on the third. The three interact, but a single purchase or a single continued use cannot establish that all three have changed at once.

A promise offers an account of future outcomes on the basis of which the recipient commits, waits, or forgoes other opportunities in the present. That fulfillment is delayed does not mean the promise is false; but the time needed to verify outcomes may not coincide with the time at which the operator collects revenue or the institution confers recognition. Who can modify the conditions of performance, and who can demand a fresh verification, thus become questions of power within intertemporal relations.

Chapter 23 distinguishes publication, research recognition, and the growth of knowledge within the production of academic knowledge; Chapter 24 distinguishes model adoption, market fit, professional recognition, and knowledge of risk within the production of financial-market knowledge. This chapter goes on to examine how organizational relations affect the conditions under which such judgments are formed and maintained. Method, habit, authority, and promise offer mechanisms that can be tracked item by item in different historical objects.

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§3 Salvation and Commercial Rule: Two Historical Comparisons That Must Not Be Merged

§3.1 Time, Relations, and Mediation in the Belief in Purgatory

Le Goff studies how Purgatory gradually took shape as a definite otherworldly space. He concentrates on changes in concepts and vocabulary in the second half of the twelfth century, while tracing earlier ideas of purification, the practice of praying for the dead, and the attendant controversies. In his account, Purgatory opened between death and the Last Judgment a situation in which change could still occur: the prayers and aid of the living could be understood as helping the dead (Le Goff, 1984).

This changed not only one image of the afterlife, but also how the living were to act. Wills provided for the disposition of property; religious communities undertook the duty of prayer; the family's concern for the dead continued to unfold through these organizations. Le Goff shows that this linkage at once strengthened the ties of family and religious community, extended the power and revenues of the Church, and lent support to the system of indulgences.

Seen from the relational structure, the dead, their living kin, the communities undertaking prayer, and the authority interpreting salvation were connected within this practice. The living's present arrangements of property and daily obligations were invested with meaning for the future condition of the departed. The future thereby refers not only to outcomes a person can enjoy while still alive; it enters relations that span individual lives. This is one historical clue to how institutions organize temporal orientation: prolonging obligations, sustaining memory, and enlarging the power of intermediaries can occur together, and need not always take the form of shortened waiting.

The comparison that suggests itself here is how intermediaries organize the relation between a person and an outcome that cannot be directly verified in the everyday course of life. What the recipient can confirm may be that a ritual was held, a donation completed, a document obtained; the relation between these observable acts and final salvation depends on a body of religious interpretation and institutional authority. Ritual performance, institutional recognition, and the final outcome are different levels of the matter.

Le Goff's analysis of this arrangement further reveals the quantitative structure that grew up inside it. The moment of a soul's deliverance depended on the quantity and quality of the sins remaining to be purged and on the intensity of the suffrages offered by the living; there thus arose in the hereafter a variable, measurable, and consequently manipulable time-scale, a system of accountancy in which quantities of sin, quantities of suffrage, and lengths of punishment were converted into one another. Alexander of Hales's reflections on proportionality gave a theoretical sketch of this conversion, which Thomas Aquinas sought to head off; the development of the system of indulgences opened the door to every sort of excess with respect to this system of accountancy. In 1300 Boniface VIII granted pilgrims who came to Rome for the jubilee a plenary indulgence—complete pardon

of all sins, a favor hitherto bestowed only on crusaders—and on Christmas Day of that year extended the benefit, unprecedentedly, to souls in Purgatory. The pope's power to free souls from Purgatory immediately had until then been a theological delineation only; thenceforth a new phase in the inclusion of indulgences in the system of Purgatory had begun (Le Goff, 1984).

This institutional chain took visible form in the German campaign of 1517: to service his debt to the Fugger bank, Albrecht, archbishop of Mainz, commissioned the Dominican friar Johann Tetzel to market the indulgence. Traveling from town to town with printed receipts and an accompanying Fugger accountant, Tetzel priced the seven years' penance owed for a mortal sin at a quarter-florin per sealed letter of remission, and with the pitch that "as soon as the coin rings in the bowl, the soul for whom it is paid will fly out of purgatory and straight to heaven" welded payment, the remission of sin, and the condition in Purgatory into a single instantaneous transaction (Manchester, 1992). This was not a commercial abuse external to the belief in Purgatory; it was the extreme to which that manipulable time-scale and system of accountancy were carried under specific conditions: obligations spanning death were finally cut into exchangeable, purchasable, independent units—and it is this chain, grown from inside the institution itself, that Chapter 23 invokes when it takes indulgences as an example of unitization. The research question that can be retained becomes more specific as well: when a single institution takes part at once in interpreting the need, organizing the performance, and certifying the qualification, where can the recipient mount a verification that does not depend on the institution's account of itself?

What this comparison reveals is a structure of mediating authority. It is limited to the mediation between performance, recognition, and the outcomes they concern; the inability to verify an otherworldly outcome directly is not the same obstacle as being unable to verify a transaction because its records are closed. When a chain of obligations spanning death is finally cut into units with a posted price, what remains for dependents to verify is often the price alone.

§3.2 The East India Company: How Private Gain and Public Rule Combined

The East India Company offers another comparison. What is at stake here is not merely the authority to interpret distant outcomes, but a commercial organization acquiring a capacity to rule that could directly alter the living conditions of others. Dalrymple records that in 1765 the Company obtained the revenue-collection rights over Bengal, Bihar, and Orissa, with revenue collection backed by its own armies. A trading organization thereby became deeply involved in local governance; commercial revenue, military expansion, and political power reinforced one another (Dalrymple, 2019).

This process cannot be described simply as the Company shaking off all state power. The Company depended on its charter, military backing, and political connections in Britain, and it also sought to use its legal distinction from the government to protect its interests. Dalrymple gives a detailed account of the relation between Parliament and the Company: private wealth could influence politics, and state power could in turn protect the Company's operations. The interweaving of Company and state was constitutive of its power.

Company rule also confronted different groups with different futures. Dalrymple recounts that while Bengal suffered famine and devastation, shareholders in London could decide to increase the dividend on the strength of revenue and the share price. What was income for shareholders could be, for local producers, the loss of the resources needed to sustain life. This difference is in the first instance a question of how income rights, information, and responsibility are connected; it must not be reduced in advance to one group's being naturally

more short-sighted (Dalrymple, 2019).

Having acquired immense power, the Company later fell into financial distress; the rescue of 1773 came with the state's strengthened supervision and constraint over it (Dalrymple, 2019). Rescue did not compensate for colonial harm, nor did it mean public control was complete; it shows at least that the expansion of private power and the state's redefinition of its conditions could occur within one and the same history.

When a structural comparison is made with platforms, the more explanatory common feature is how an organization operating in the name of business acquired the capacity to set the conditions of other people's activity, and how the attribution of gains and the bearing of consequences came apart. Platforms do exercise private governance when they set the terms of content access, transaction eligibility, and account disposition. But an account suspension is not a colonial conquest, and a platform commission cannot be judged a tax merely because it is levied proportionally. The legal sources, instruments of coercion, populations concerned, and conditions of redress of the two kinds of power must be analyzed separately.

This difference further requires avoiding the move of writing the older rule as wholly visible and the newer rule as wholly invisible. Colonial rule likewise had its complex intermediaries, justifying narratives, and internal differences of information; platform users may likewise know full well that they are subject to rules. Power need not always be invisible to be effective. Knowing who makes the rules and having the power to change them are two different things.

§3.3 The Uses and Limits of Historical Comparison

The belief in Purgatory and the practices of salvation surrounding it show how obligations spanning individual lives were sustained through organizations—and how, in certain periods, they were converted into purchasable units; the East India Company shows how commercial gains and the consequences of rule were distributed across different groups. The former could lengthen the time horizon of practice; the latter could detach one party's present income from another party's long-term living conditions. They illuminate different relations, and they cannot be merged into one and the same kind of "early platform."

Table 2. What the two historical comparisons retain

Historical object	The questions it can serve in contemporary analysis	Where the comparison must stop
The belief in Purgatory and the associated practices of salvation	How intermediaries connect the living and the dead, present performance and future condition; how obligations and remission came to be converted into quantifiable, purchasable units	The cross-era comparison from Purgatory to platforms cannot establish transmission by similarity; religious relations are not financial transactions, and doctrinal recognition is not algorithmic scoring

Historical object	The questions it can serve in contemporary analysis	Where the comparison must stop
The commercial rule of the East India Company	How commercial interests acquired rule-making and coercive capacity; how one party's income and another's long-term living conditions came apart	Colonial conquest is not service dependency, and armed tax collection is not a platform fee; the differences in forms of coercion and legal relations must not be erased

The two comparisons share one further limitation: they cannot substitute for the actual history of modern media and advertising. To understand why platforms could organize attention, content, and commercial revenue through continuing services, one must enter the more concrete operations of media business.

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§4 The Commercialization of Advertising: The Organization of Promises, Distribution, and Audiences

§4.1 The Competition of Promises as Johnson Saw It

In 1759, in number 40 of *The Idler*, Samuel Johnson discussed newspaper advertising. Advertisements, he wrote, had become so numerous that readers no longer perused them attentively; advertisers therefore competed for attention with grander promises and more stirring language. He then summed up this competition in one brief sentence:

> Promise, large promise, is the soul of an advertisement. (Johnson, 1759)

Johnson lists advertisements for wash-balls, bed-coverings, and beauty products, and discusses appeals that play on a mother's anxiety for her infant's safety. His criticism is not aimed only at obvious puffery: some advertisements deliberately display restraint, conceding that the product cannot do a certain thing, so that the remaining promises gain credibility. This is a contemporary observer's identification of modes of persuasion, not a laboratory report that all these products were ineffective.

The material supports one limited but important historical judgment: in the commercial environment of the eighteenth-century newspaper, competition for attention, the inflation of promises, and the appearance of credibility were already being discussed together. It does not prove that all advertising of the day went unfulfilled, nor that modern platforms merely accelerated the same technique. What Johnson saw was commercial expression in a particular print environment; the organizational forms that came later still require separate account.

The ending of the essay carries a further worry, one that bears directly on the production of knowledge. Johnson thinks of those in later ages who will preserve and read these newspapers: confronted with many merchants each claiming their product or craft to be the best, how will posterity reconcile these conflicting records? Once commercial expression enters the archive, it may be treated as material for knowing the past. It does preserve history—but what it preserves, first of all, are the claims publicly made at the time, not proof that those claims were honored.

Advertising thus affects not only immediate purchases; it also takes part in forming the world that can later be read. The historian needs to identify the purpose for which material was produced, just as the contemporary user needs to identify the conditions of selection behind a platform's outputs. Human writing is real activity, but it is not therefore a record unselected by interests and institutions of transmission.

§4.2 How Brand Advertising Changed the Balance of Distribution

The effect of advertising goes beyond competing for readers' attention. Vincent P. Norris studied how American advertising textbooks explained the rise of national advertising, criticizing them for presenting the process chiefly as manufacturers conveying product information to distant consumers. The industry literature he assembles suggests that manufacturers also hoped, through brand demand, to change the balance of power between themselves and the wholesalers and retailers (Norris, 1980).

The Calkins case as related by Norris makes the mechanism clear: once consumers persistently demand a certain brand, retailers need to stock it; if the incumbent wholesaler cannot supply it, the retailer has reason to look for other channels. Advertising did not make the work of distribution vanish; it could change who had the standing to set terms for the distributors. Calkins's account, relayed by Norris, shows how a practitioner understood this strategy; it does not by itself establish its effects across the industry.

Laird's historical study of nineteenth-century American business and advertising unfolds the question through the relations among manufacturers, retailers, and the professionalization of advertising. She shows that manufacturers deliberately developed advertising to acquire market control previously held by wholesalers and retailers; at the same time, the multiplication of new products made introducing a product's uses and building demand for it an actual business task (Laird, 1998). Informing the consumer and contending for power are not either/or. Advertising could make a new commodity known, and it could also give a brand a stronger position in channel bargaining.

A transmission across links thereby took shape: the manufacturer buys opportunities of communication; communication shapes consumers' recognition of, and demand for, the brand; that demand in turn changes the options of retailers and wholesalers. Advertising expenditure acts upon audiences, yet it may be realized in distribution contracts and price negotiations. The consumer is no longer only the recipient of whatever a given retailer chooses to stock; carrying brand demands formed elsewhere, the consumer can in turn influence the retailer. The direction of return in the relation changed, and power was redistributed with it.

Branding also allowed expectations of a product to persist beyond any single face-to-face transaction. This can lower the cost of repeated judgment, and it also makes consumers depend more on what they know of the brand's producer. How advertising content, prior use, and merchants' recommendations jointly support this belief must be established separately; brand demand can be presumed neither naturally given nor uniformly fabricated.

What this history offers the analysis of platforms is not a law that "advertising naturally monopolizes," but a relation worth tracking: how the capacity to reach consumers changes the conditions of choice for other operators. In the digital environment, manufacturers may still use brands to strengthen their bargaining position, yet distribution platforms may also control the paths by which brands reach consumers. One cannot presume that once the old intermediaries vanished, all power returned naturally to producers or consumers.

§4.3 The Images of Life in Advertising Are Not a Direct Mirror of Society

In his study of American advertising from 1920 to 1940, Marchand terms the many advertisements that depict human relations and living settings social tableaux. These images display the family, work, and sociability; they look like a slice of everyday life, yet they serve, first of all, the sale of goods. The class, housing, and ways of associating shown in the frame are selected by marketing strategy, conventions of representation, and the advertisers' own horizons (Marchand, 1985).

This selection cannot be explained simply as advertising falsifying reality altogether. An advertisement needs its audience to recognize a situation before it can connect the commodity to that situation; but the audience's ability to recognize does not make the picture socially representative. Marchand adds that one must not swing to the opposite extreme and read advertisements as mirrors of consumers' actual wishes: they contain both the audience's expectations and what the makers thought the audience ought to aspire to.

Here the promise expands from single-item performance to relations of life. A commodity can be presented as the means to respect, to holding a family together, to avoiding embarrassment, or to entering a certain kind of life. Goods may genuinely improve life; yet between the purchase of a product and these relational outcomes there still intervene other people's actions, particular circumstances, and sustained effort. Buying an item does not guarantee the respect, the family bonds, or the social acceptance that the advertisement promises.

The distinction drawn in Chapter 1 between working in time and working on clocks concerns precisely such sustained processes, which cannot be certified by the receipt for a single purchase. Family, occupation, and social recognition are not completed by the purchase of an object; human labor and relations must continue in time. Advertising can place a purchasable object in the foreground of such processes, while the conditions needed to maintain them recede into the background. Whether this substitution has occurred must be read from particular advertisements and their contexts of use; a commercial source alone cannot decide it.

The same body of material can therefore serve different tasks. To study how advertising imagined society, one can analyze advertisements directly; to study how families actually lived at the time, other records are needed; to study whether advertising changed life, evidence of diffusion, reception, and behavior is required. What the fiber denotes is precisely the different states that remain compatible with a given reading: the same family advertisement may be compatible with a widely existing way of life, or with an aspiration most people had not yet realized. The picture alone cannot rule out the latter state, still less can the circulation of the advertisement establish that this life had already become general.

§4.4 Why Media Face Audiences and Advertisers at Once

Advertising at scale also required a business system that continuously organized audiences. Media produce or aggregate content; readers and viewers arrive for the content; advertisers pay for access to these audiences. The relations among audience size, composition, mode of reach, and advertising price make content production and commercial operation mutually contingent. Advertising does not merely happen to appear beside a finished work; it can take part in deciding the conditions under which works continue to be produced.

Smythe's audience commodity thesis demands that communication research analyze not only information and meaning, but also the audiences that advertising-supported media produce and sell. He places watching and consuming within capitalist production and the reproduction of labor (Smythe, 1977). This is an interpretation with an explicit theoretical position; it cannot, without argument, convert all reading and leisure directly into

labor. But it forces the researcher to see that the payer and the user may not be the same party, and that what the medium draws from use is not only a stream of subscription revenue.

Two-sided market theory explains the arrangement from pricing and mutual dependence. Rochet and Tirole show that a platform must bring both sides on board at once, and that the choice is not only the total price but also how it is distributed across the two sides; one side may be subsidized while the other bears the main charge (Rochet & Tirole, 2003). This explains how low prices or free access for audiences could become a viable business; it cannot by itself judge whether the arrangement is fair to the participants.

The audience commodity thesis and two-sided market analysis do not pose the same question. The former tracks labor, commodification, and relations of value; the latter studies participation and price structure. This chapter retains that difference in using them: one cannot infer from one side's being free that no exchange exists, nor infer from the existence of exchange that users are necessarily being deceived. What must continue to be ascertained is what the medium promised, what it actually delivered, how the audience's activity became the basis of revenue, and who can change these arrangements.

The history of advertising thus appears as a reorganization of multiple relations: manufacturers could form brand demand beyond the established channels; media turned access to audiences into advertising revenue; professionals in turn placed commodities into images of life. These arrangements determine not only who receives income; they also shape how consumers understand a life that has not yet arrived.

§4.5 Installment Buying and Model Changes: How Waiting Acquired New Meanings

Marchand's study of consumer ethics makes the relation between advertising and time preference concrete. The automobile competition of the 1920s involved not only which car was cheaper or more durable, but also styling, installment buying, and when a car ought to be replaced. The technical lifetime of the good, its newness or datedness in social evaluation, and the timing of payment could cease to coincide (Marchand, 1985).

He records that in its 1926 advertising Ford still appealed to living within one's means and to the needs of transportation, setting itself against the consumption oriented toward styling. When the new car was introduced in 1927, the advertising agency tried to shape the new image through refined social tableaux; Ford himself still resisted the logic of continual model changes, claiming that he wanted the car to prove so durable that no one need ever buy a second. This is not a history in which all operators simultaneously accepted one consumer creed. Visible conflicts ran among business competition, the advertising trade, and personal conviction.

Installment buying further changed the temporal arrangement of action. The consumer could take the goods first and pay in installments later; closing a sale no longer necessarily required the full price to be accumulated in advance. Marchand also recounts how the banking world gradually reinterpreted installment buying, once morally suspect, as a form of thrift. What changed at once were the feasible arrangements and the language of evaluation: buying first became technically and contractually possible, and in certain commercial discourses it gained legitimacy.

These changes cannot be explained by "consumers becoming less patient" alone. A person may still prize a long-term life yet obtain needed goods earlier because credit conditions changed; may replace an older model earlier because continued use is expected to carry a social cost; and may, through repeated exposure, genuinely have revised his evaluation of new and old and of worthiness. The same earlier purchase is compatible with different time preferences, expectations of consequences, and payment conditions.

Advertising in the Great Depression offered yet another attempt at organizing the future. Marchand records that businesses and the advertising trade urged present purchase in the name of restoring confidence, against hoarding, and in support of prosperity. But the repeated announcement that recovery was just ahead also met public skepticism and satire; inspirational advertising also served the business community's need to sustain its own confidence (Marchand, 1985). Persuaders did not only try to change consumers; they may also have consolidated their own judgment in the optimistic formulas they exchanged with one another.

Long before digital platforms, businesses were already organizing a picture of the future, present action, and continuing payment together, so that the meaning of waiting changed. It does not prove that propaganda had already changed everyone's time preference; it does identify a mechanism that needs tracking. The content of advertising, the credit contract, and the cycle of renewal change choice together; from sales records alone one cannot tell which link did the work.

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§5 Mass Communication and Behavior Design: From Organizing Opinion to Arranging Interaction

§5.1 Bernays's Claim as a Self-Description of His Profession

In *Propaganda* (1928), Bernays treats the conscious organization of the public's habits and opinions as an important activity of democratic society. He opens the book with these words:

> The conscious and intelligent manipulation of the organized habits and opinions of the masses is an important element in democratic society. (Bernays, 1928)

He then connects this work to the complexity of society: no one can personally investigate every public question or test every commodity before buying it, and people therefore depend on organizations that sift information, foreground problems, and narrow the range of choice. In his account, propaganda is not merely an instrument of manipulation; it is a necessary intermediation by which a mass society can function at all (Bernays, 1928).

This argument cannot be taken simply as proof that an omnipotent behind-the-scenes government already exists. Bernays is at once describing society and seeking standing for his own profession. He moves from "people need intermediaries" to "certain professional organizations are qualified to perform this intermediation," and then uses the necessity of organizing opinion to justify propaganda. These several judgments need to be examined separately. To grant that people cannot verify everything is not yet to grant who is to do the sifting, toward what ends, and how those affected may object.

The connection between this historical material and the definition of this chapter lies not in Bernays having foreseen algorithms, but in his explicit assertion of a claim to governance: organizing people's information and choices can proceed while formal individual liberty remains intact. Platforms later took over part of this work of sifting within everyday services; but the fact that both are engaged in sifting cannot by itself establish a direct inheritance among the modes of organization.

§5.2 Packard Recorded a Body of Persuasion Knowledge in Commercial Operation

Packard's *The Hidden Persuaders* observes commercial persuasion from the other side. He discusses how advertising, public relations, and related fields borrowed knowledge from psychology and psychiatry in search of

the causes behind consumption habits, preferences, and reactions, and put this knowledge to work in marketing. What concerns him is no longer only what advertisements say, but why corporations pay to study people, and how research enters production and communication (Packard, 1957).

Within this mode of business, surveys, interviews, psychological interpretation, creative production, and the scheduling of communication were divided into tasks that professionals could perform in collaboration. A judgment about some desire or anxiety no longer had to remain attached to a salesman's intuition; it could be recorded, taught, and applied to the next project.

Yet a method's being sold, being adopted, and being reported as effective are not the same evidence. Packard's book is a contemporary critical investigation; its record of the industry's efforts and of researchers' claims cannot be converted unconditionally into experimental confirmation of the efficacy of every technique. That the advertising trade believed it could reach deep into people is worth studying; under what conditions it actually changed behavior is a further question requiring evidence.

This point bears directly on how the history of platforms should be written. Psychology, psychoanalysis, behaviorism, and later product design cannot be fused into one engineering pipeline that had already mastered the human mind. The several traditions studied different objects with different methods, and their applications could fail. Industrialization means, first, that research and deployment acquired sustained organizational conditions; it does not mean that human responses had become arbitrarily controllable variables.

§5.3 How Behavior Design Intervenes in the Conditions of Action

In his 2009 paper, Fogg attributes the occurrence of a target behavior to the conjunction of motivation, ability, and a trigger at the same moment. The paper states that his model of persuasion targets behavior rather than attitudes, and that its axes are a conceptual frame for expressing relations, not a quantitative relation with precise units (Fogg, 2009). Designers can thus check separately whether a person is willing, whether he can complete the action, and whether the prompt arrives at the right moment, rather than attributing every failure to insufficient motivation.

His examples concern how a website induces visitors to submit an email address. If the form poses a hard problem, a willing visitor may still be unable to complete it; if the behavior is made simple and prompted at the right moment, the likelihood of completion changes. The operative focus shifts from persuasive discourse to the conditions of behavior: without first changing a person's beliefs every time, one can change actual conduct through the difficulty of the task and the timing of the prompt.

Such design delivers real functional gains. A clear, easy form can help a person do what he already meant to do; a well-timed reminder can support a goal he set himself. To what ends a given body of design knowledge is put remains a separate question. More registrations show, in the first instance, only that the target behavior occurs more often; they do not by themselves prove that registration serves the user's long-term intentions.

When records of interaction can feed into the next round of design, operators acquire not only richer information about audiences but also the opportunity to go on changing the environment of exposure. Interface, content, prompts, and the results of action can feed back into one another. This possibility, however, must be distinguished from actual deployment: that Fogg proposed a model does not mean that all platforms adopted it at the same time, still less that every feedback loop aims at eroding judgment.

Mass media had long studied their audiences and adjusted programs and advertising in response. Digital interaction allows recording, prompting, and subsequent presentation to be linked more tightly. If the next prompt appears the moment an action has ended, the designer influences not only a given choice but also the interval between choices. To stop, a user may have to interrupt an ongoing chain of activity, rather than wait for the previous activity to end of its own accord.

Two different kinds of return also exist here. A user's stating a reason that obliges the designer to reconsider the objective is a response among reasons; a user's staying or leaving, used by the system to raise the attainment rate of a fixed objective, is behavioral feedback. The latter too can carry valuable information, but two-way transmission of data does not automatically constitute the former relation. Applying this distinction to platforms is this chapter's analysis; Fogg's behavior model itself does not adjudicate what rights participants hold.

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§6 How Relational Structures Change Belief and Intertemporal Choice

§6.1 Accessing the Same Information and Knowing That Others Have Accessed It Are Not the Same

In studying common knowledge, Chwe attends to a special condition in coordinated action. That several people each receive the same message is not necessarily enough for coordination: each may also need to know that the others received it, and to know that they know. In the formal definition, everyone in the group knows the matter, everyone knows that every member knows it, and this mutual knowing continues without end. This does not require people actually to run through infinite levels before acting; public rituals, assemblies, and media events can supply the conditions under which mutual expectations form in practice (Chwe, 2001).

Common knowledge is not unanimity of opinion. People can know that the same announcement has been made public while disagreeing with its claims; they can each approve of a practice yet not know whether the others approve, and so be unable to coordinate. Sharing material, believing the material to be true, and endorsing the objective of an action are different relations.

Chwe uses mass media and advertising to show that the value of communication sometimes lies not only in how many individuals it reaches. For goods or activities that become more valuable when others take part, one widely known public transmission can lead recipients to form expectations about the other recipients. He also stresses that ritual can sustain existing authority and can assist political change alike; common knowledge is not the property of any one order.

Chwe further uses social networks to show that spreading faster and coordinating better need not coincide: in networks of strong ties, where friends of friends know one another, information has difficulty leaving the original circle, yet a person can more easily judge what the others know; networks dominated by weak ties among acquaintances are better at carrying news further. This comparison gives topology a definite object of explanation: what changes is not the truth or falsity of a message, but the conditions under which participants form judgments about what others know (Chwe, 2001).

This makes the publicness of platforms an independent question. A system can be very good at delivering content to each individual yet not equally good at letting participants know what the others have encountered. Conversely, a public record of modest scale may support common verification because its content is stable and its conditions of access clear. Efficiency of transmission cannot vouch for the capacity for common verification.

§6.2 Which Conditions of Verification Personalization Changes

Personalized distribution can lower the cost of finding relevant material, and it can also cause different people to see different versions of the same information. The latter change is not by itself sufficient proof of harm. Experts and novices need different explanations; local and national information need not be identical; common life has never required everyone to go on reading the same text.

The difficulty appears under other conditions: participants cannot know whether the same matter has been presented differently to others; versions lack a stable record; and the rules of display change with each individual's reactions. What each person sees may then be relevant to its recipient's circumstances while lacking a stable version that others can inspect. If a transaction promise, a rule's explanation, or a piece of public information can appear only briefly in a personal interface, the dispute concerns not only the truth of the content, but also who saw what at the time, and whether that fact can later be recovered.

A stable public record provides an object for joint verification, but it does not by itself create common knowledge; participants must also be able to judge whether others have encountered and understood the relevant information. The significance of a public observation interface therefore lies not in manufacturing uniform opinion, but in allowing disagreement to be directed at objects that can be jointly identified. It can be composed of stable announcements, version records, public reasons, and traceable material; it need not be a traditional newspaper or broadcast. The common visibility of traditional media guaranteed neither fairness nor truth, and the differentiation of digital media does not preclude public records and collective collaboration.

From here one cannot proceed directly to the conclusion that personalization causes social polarization. Differences in content encountered, judgments about what others know, changes in attitude, and enduring political hostility require different kinds of evidence. If all results pointing in different directions are read as the dissolution of common knowledge, the framework loses the possibility of being corrected by counterexamples.

§6.3 Fiber Analysis: What a Given Reading Actually Rules Out

The fiber framework of Chapter 4 analyzes observation as follows: under a given mapping, which true states remain compatible with one and the same reading. In actual research the mapping is usually only partially known, so explanations not yet excluded by the evidence must be retained on the basis of existing knowledge. This tool does not demand every dimension of reality; it demands only that the observer, the task, and the conditions of judgment first be fixed.

Take a report on advertising effectiveness. The following is an analytical scenario, not a transcript of any experimental result. The report accurately records a set of clicks and the purchases that followed. For the advertiser, the task may be to identify how many additional purchases the advertising produced; for the publisher, to identify how advertising revenue is distributed; for the user, to understand why this advertisement was received. The three parties see certain identical numbers, but their tasks of judgment, the material available to them, and their conditions of knowledge differ; characterized by the identification set under partial knowledge of the mapping function, as set out in Chapter 4, the explanations the three parties need to distinguish are not the same—they do not in fact face the same object of analysis.

For the advertiser, the record of real purchases may still include purchases that would have occurred anyway, and purchases that the advertising actually added. Increasing the precision of the logs cannot by itself distinguish these two explanations. A suitable controlled design can add to knowledge of the incremental effect, but such

confirmation answers only the advertiser's task; if the publisher asks further whether the distribution of revenue is fair, and the user asks further whether the use of the data accords with his own authorization, each requires its own standards and evidence, and neither follows from advertising effectiveness.

For the publisher, the settled total does not yet show the fees and rule effects at each link. Caliskan, MacKenzie, and McGowan record that the different auction logs publishers obtained from Google were criticized as difficult to verify against one another; the platform also took part in ad delivery, reporting, collection, and rule-setting (Caliskan, MacKenzie & McGowan, 2026). Verification is thus a question not only of information but of the rights to obtain information and to interpret the rules. Providing a clearer report may help, yet it does not automatically change which trading conditions the publisher can choose among.

The **pseudo-small fiber** occurs when, while important compatible explanations remain, the observer is led to a definite judgment unsupported by the evidence. To take the association of clicks and purchases directly for the incremental effect of the advertising, for example, is to exclude in advance explanations that have not yet been tested. But if the advertiser knows the report's limits perfectly well and still must use it because of business arrangements, the problem ought no longer to be described as his simply mistaking the number for truth. What must then be asked is why other judgments cannot enter the transactions and the allocation of resources.

The example also distinguishes three kinds of improvement. Correcting a log error improves the reading; establishing a suitable independent control increases the identification of a given effect; changing rights of access to transactions and data, or the right to appeal, changes the actual conditions of action. The three can complement one another, but the completion of the first is no reason to declare the other two solved. What fiber analysis contributes is the requirement that a claim of greater transparency specify which judgment, which evidence, and which condition of action has improved.

§6.4 When Feedback Aids Cognition and When It Replaces It

Platforms record users' reactions, use these reactions to adjust content and interface, and then record the reactions that follow the adjustment. Content producers likewise adjust production in the light of visibility and revenue. The data so formed contain both people's existing interests and the influence of display, prompts, business choices, and producers' adaptations. To treat them wholesale as an unmediated sample of preferences is to overlook the platform's own part in forming the sample.

Feedback has an actual learning function. If a system revises its arrangements in the light of task failures and thereby makes it easier for users to achieve their own purposes, this return can improve the tool. Users' own purposes here do not mean purposes untouched by social influence, but purposes whose conditions users have an opportunity to understand, which they can compare with alternatives, and which they can endorse or revise accordingly. Chapter 17's distinction between functional and pathological recursion warns the analyst not to pronounce pathology from the mere existence of a loop. What must be tracked is which output returns to which input, what changes after the return, and what independent basis remains able to correct the process.

The risk appears in a chain like this: the platform allocates visibility by some participation metric; producers adjust their content accordingly; the reactions to the adjusted content are then taken as proof that this allocation fully meets the need. If that proof no longer requires checking the results of the task, the quality of information, or purposes the users can continue to endorse, the system may chiefly be using the activity it has itself shaped to confirm its own correctness. More feedback then does not mean more independent grounds.

But even if a given piece of content is proven to have an incremental effect, the overall arrangement cannot be inferred to be legitimate. An advertisement can effectively bring about purchases; a recommendation can effectively lengthen reading; both may achieve business objectives; whether they improved the user's life, whether they left him more able to judge and to act, involves another set of outcomes. The extension of a genuine local achievement into a broader guarantee of competence or legitimacy is the narrative leap that here needs auditing.

The Macro-Cognitive Accounting Framework (MCAF) of Chapter 3 attends to how such leaps accumulate across time, and to who bears the costs of verification and of failure. **Meta-audit capacity** here is not a matter of reading a few more platform reports; it is the capacity to keep verifying the grounds of reports, to question the objectives of evaluation, and to press for changes in the conditions of use. It requires not only expertise but also organization, time, access to data, and authority to act—knowing about a problem does not mean these conditions are in place.

Nor can human life be fully summarized by these tasks of verification. Producing content, learning a profession, and maintaining relationships all require investment without immediate metric returns. If a person's income and visibility come to depend ever more on short-term feedback, then even knowing that the score does not equal the worth of the work, he may find it hard to preserve the rhythm of long-term work. What is lost here is not a piece of information; it may be the conditions for forming judgment, sustaining relationships, and bearing long-term obligations. Whether this loss has occurred must be observed in actual work and life, not counted from interactions on a screen.

Advertising revenue, time spent, and market capitalization therefore cannot serve directly as measures of cognitive debt. It requires a trackable process: how commitments exceeding the support of the existing evidence entered action; how independent judgment and the conditions of correction were impaired; and how the consequences were deferred and transferred to others. If these conditions do not hold, business growth and frequent use may still be only the expansion of a useful service.

§6.5 Changes in Belief: Materials, Habits, and Expectations about Others

A relational structure can first change the materials accessible to the formation of belief. Search and ranking determine which explanations come more easily into view; textbooks and prevailing formats determine which inferences are more easily recognized as complete. That material has been organized does not make it false; the question is whether participants, when they form judgments on the basis of it, can still identify the conditions of selection, and the other explanations that have not come into view.

Another path runs through repeated action. The habit Pascal discusses lets reasons once accepted be maintained without being rebuilt from the ground up each time; in the environment of platforms, a practice repeatedly used and rewarded may likewise gradually become participants' common sense about how things are done. The correspondence here is a mechanism hypothesis that needs testing: have users moved from "this practice works here" to "this is how things ought to be done"? To support the latter judgment requires material on their evaluations, explanations, and actions across settings—frequency of use cannot stand in for it.

Belief also forms through expectations about others. A commodity, a claim, or an activity may be worth joining in not only because the individual likes it, but because he believes others will also endorse it. Public rankings, displays, and discussion can support such mutual expectations; personalized presentation can also lead people to misjudge how widespread something is. Chwe's analysis shows that public communication has its power

precisely because it can change this higher-level belief; this differs from persuading each person separately of the same sentence.

When a platform both arranges what material appears and selects which reactions are visible, and then uses these reactions to adjust the next round of the environment, the several paths can become connected. A participant sees a filtered set of choices and reacts; that reaction becomes material for later arrivals judging what is popular, and also the basis on which the platform goes on selecting. Without fabricating every item of content itself, the controller can take part in organizing the conditions under which some beliefs are more easily formed and maintained.

This relation of return cannot be evaluated by the frequency of feedback alone. If dissent, failure, and purposes that users newly propose can change the rules of selection, the return can extend knowledge; if this information is only converted into parameters that sustain participation, correction by reasons may give way to the regulation of behavior. The difference lies not in whether the system answers, but in whether participants can bring about change in the question itself and in its criteria of evaluation.

§6.6 Time Preference, the Credibility of the Future, and the Rhythm of Action

This chapter uses time preference for the relative weight a person assigns to earlier and later gains or losses when the nature of the outcomes, their probability of realization, and the constraints on action are comparable; actual intertemporal choice is also affected by payment conditions, expectations of fulfillment, and windows of opportunity, and cannot be read directly as a measurement of time preference. The speed at which messages arrive, the frequency of screen switching, and the cycle of rule updates are only features of the environment of choice. The historical materials can show how certain arrangements organized waiting, earlier consumption, or sustained practice; they have not measured participants' discount parameters.

First, structure can change the conditions that waiting requires. Installment buying separates obtaining the good from completing payment; long-term guarantees of eligibility give an investment a better chance of continuing; rules that can be withdrawn at any moment add to future rewards an uncertainty controlled by others. A person may still prize long-term outcomes yet choose the reward redeemable now, because he cannot sustain his life or fears that eligibility will lapse. This is a change in the conditions of action and in the credibility of the future, and it should not be called directly a corruption of preference.

Second, structure can change the meaning of waiting. Model changes make goods still usable appear dated; public comparison makes future social approval a reason for present purchase; religious practice can interpret present restraint as fidelity to a more distant goal. These do not all shorten the time horizon. The change occurs in what outcomes are considered worth pursuing, and in how the relation between present action and future outcomes is understood. What changes first is the value of outcomes and their social meaning; this cannot be equated directly with a strengthening preference for immediate reward.

Finally, repeated practice may change the rhythm in which a person habitually acts. If every hesitation means losing a visible opportunity, while immediate response is continually rewarded, participants may gradually give up the activities they once used for comparing and waiting. The step from here to a change in stable time preference still requires further evidence: whether, once the old regime of reward and punishment is left behind, the intertemporal trade-off persists; whether the participant still endorses it; whether choices change once the conditions of waiting are restored.

Table 3. The same short-run choice may involve different mechanisms

Observed behavior	Explanations that still need to be distinguished	Material that increases identification
Buying earlier	A stronger preference for immediate gratification; changed payment conditions; expectations that the good or its price will change	Intertemporal choices under comparable cost and availability; contract terms; judgments about the future
Shortening the production cycle	Greater weight on immediate metrics; tightened income constraints; fear that distribution or eligibility will be withdrawn	Choices before and after stable support appears; production plans; records of rule changes and payment
Repeatedly responding at once	A settled endorsement of this rhythm; the arrangement of prompts; a strengthened belief that opportunities will be missed	Behavior after prompts and windows of opportunity change; evaluations and reasons after delay

These distinctions let the relational analysis explain an apparent contradiction: agents who still prize a long-term life make more and more short-term choices. The reason may lie not in some overall decay that first took place within, but in the paths by which long-term goals are realized having been changed. If this situation persists, beliefs, habits, and preferences may in turn be reorganized, forming a deeper dependence.

The power of the Discrete Leviathan therefore lies not only in offering more immediate temptations. It may at the same time alter whether long-term goals remain attainable, whether future rewards are credible, and how quickly people must respond in the present. Testing this power requires bringing together people's explanations and experience, actual resources and rules, and the variation of repeated choices; no single choice or metric can establish how a person's time preference or long-term arrangements have changed.

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§7 The Practical Boundaries of State, Capital, and Private Governance

§7.1 A Trans-Sovereign Network Is Not a Stateless Network

Operating across borders raises the difficulty of coordination in governance, but it cannot turn states into naturally impotent bystanders. The establishment of nodes, the enforcement of contracts, access to communications, and the arrangements of investment all take place within an institutional environment. The state may restrain private control, and it may also support it through licensing, procurement, infrastructure arrangements, and industrial policy. The relations of support, restriction, and dependence between state and platform must be judged by concrete policies, organizations, and transactions; they cannot be settled from the two names "state" and "platform" alone.

Schiller's study of digital capitalism places the expansion of corporate networks within the long-term transformation of American communications policy. He records how corporate users fought to build and use their own

dedicated networks, and analyzes how the regulators drew the line between computing services and telecommunications services, moving ever more network activity toward the more liberalized side (Schiller, 1999). This history shows that the space of network operation was created not by technology alone, but also by regulatory classification and public decision. Liberalization changed the rules; it did not abolish them. What must be ascertained is which constraints were weakened, and which rights and business capabilities were extended.

The trans-sovereign character of the Discrete Leviathan therefore means that chains of control and scopes of jurisdiction do not fully coincide. A state can reach local assets, gateways, or business conduct, yet may not by itself obtain the information and the conditions of continuous enforcement that the whole chain requires. This incomplete coincidence does not mean that all interventions are ineffective, still less that one must first possess a unified global power before any local arrangement can be improved.

§7.2 Transparency, Countermeasures, and Changing the Rules Are Not the Same Step

Ananny and Crawford criticize the transparency ideal that equates seeing a system with understanding and governing it. They require that algorithms be observed within relations jointly constituted by people and technology, not merely by opening an isolated black box (Ananny & Crawford, 2018). For a cross-layer network, this means that after code or logs are disclosed, it must still be ascertained who can interpret, contest, and change the conditions that produce the results.

Header bidding provides a case in which control relations were changed at a particular stage: an arrangement by which publishers introduce multiple bidders on the webpage, allowing demand outside Google to compete at a particular link. Caliskan, MacKenzie, and McGowan record that this change strengthened publishers' control over the sale of advertising and forced Google to respond; Google then offered an alternative arrangement running within its own infrastructure, bringing part of the bidding, the data, and the eligibility conditions back under its rules (Caliskan, MacKenzie & McGowan, 2026).

The technical and collective action of publishers did change the conditions of competition; whether the choice endures depends on the rules of bidding, data, and access. The analysis should compare item by item which options increased, how long they lasted, and which dependencies remain—without issuing a blanket verdict of success or failure in counteraction.

The same study also shows that the materials, testimony, and expert reports produced in antitrust litigation gave researchers access to auction paths and internal decision-making information beyond the technical manuals. What can be confirmed here is that the conditions of discovery improved; the public release of material cannot be equated with the completion of governance. Whether the rules changed, whether the changes can be enforced, and whether those affected can keep using the new options remain practical questions for the time that follows.

In the analysis of this chapter, governance comprises at least several distinct tasks: understanding the relations, obtaining the rights to act, and maintaining alternative arrangements. An effective disclosure measure need not be dismissed for failing to complete all of them; but neither may it, on the ground of completed disclosure, declare the remaining tasks accomplished.

§7.3 The Disagreement between Technofeudal and Capitalist Explanations

Varoufakis describes with technofeudalism a shift in power: those who hold cloud capital control the gateways to digital activity, placing traditional capitalist operators and ordinary users in positions of dependence. His

claim is not that markets, profit, and wage labor have vanished from reality, but that markets and profit have lost their former commanding position, and the rent generated by platform access has acquired a new centrality (Varoufakis, 2023).

The strength of this explanation is that it moves the discussion from how much free convenience consumers receive to how operators and users are subordinated to the owners of gateways. Yet from a platform's being able to charge fees to rent's having replaced profit, argument is still required. Technical services, infrastructure investment, the bearing of risk, and market power may all enter into one and the same revenue. A high fee ratio can raise the question of market power; it does not by itself prove that the corresponding services make no productive contribution at all.

Gane opposes the judgment that capitalism has been replaced by feudalism. In his view, what needs explaining is what new forms capitalism has taken, and why these forms are increasingly hard to constrain effectively through the nation-state and social-democratic politics; calling it capitalism does not mean denying its problems of power and exploitation (Gane, 2025). What the two sides dispute is not only a name, but also how markets, profit, the state, and paths of action should be understood.

This controversy requires separating the nature of revenue, business relations, and institutional form. When a platform provides technical services while controlling the gateway to transactions, which parts of its fees rest on service inputs and which are sustained by a gateway hard to substitute cannot be read off from the total or the rate. Nor can the analysis of one business be extended directly to the whole node network. Only by continuing to compare how service gains and control gains are combined, at which links competition occurs, and whether users can change their dependence, can one judge which theory better explains these relations.

The Discrete Leviathan can enter into conversation with both explanations, but it presupposes neither as the victor. Even if the platforms still stand within capitalism, cross-layer private governance remains a real problem; even if some revenues are unmistakably rent, it does not follow that all nodes have ceased producing or no longer bear business risk. What this chapter needs to retain is the relation between rule-making power and continuing dependence, and how that relation acts upon knowledge and judgment.

Capital structure reflexivity occupies an important place here. Expected revenue can support investment in infrastructure and gateways, and that investment may in turn consolidate the conditions of future revenue; but whether this loop has over-expanded must be judged against actual operations, financing obligations, and alternative paths. Market capitalization is not a direct measurement of power, and capital expenditure is not a direct measurement of cognitive debt. One need not prove that attention has replaced the interest rate as the anchor of valuation in order to identify this network of control.

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§8 The Specificity of the Contemporary Assemblage and the Maintenance of the Capacity for Judgment

§8.1 Returning from Historical Comparison to the Concrete Assemblage

Arthur's analysis of technological combination provides a perspective on how contemporary systems connect existing technologies (Arthur, 2009). This chapter uses that perspective to analyze their composition, not to infer historical transmission.

The historical materials above reveal different ways of organizing the transmission of knowledge, the maintenance of belief, the distribution of consequences, and the meaning of waiting; they did not form successive versions of one and the same machine, yet they show that human time and belief were not always already complete, prior to organizational relations.

The specificity of the contemporary assemblage lies in this: controllers can connect the selection of materials, the allocation of opportunities, the recording of behavior, and subsequent intervention more tightly. A participant's reactions no longer merely end one act of transmission; they become the input that changes the next exposure; business revenue in turn supports the continuing operation of this arrangement. The power to organize information and the power to organize consequences thus support each other, bearing upon people's beliefs and upon the actual conditions those beliefs confront.

Generative systems further bring part of the production of expression into the service process, and systems with execution capacity can, once the corresponding authorization is obtained, complete certain operations directly. Data collection, content selection and generation, and action execution can be integrated into the same service process; the later stages need not replace the earlier ones. How long a single authorization can cover, whether comparison remains possible before execution, and who remains responsible after the results—these are the concrete temporal and rights conditions of this assemblage.

Above all, the difference between recommender systems and generative systems must not be written as one in which only options were affected before and thinking itself is affected only afterward. Distribution and reward could always change what human beings are willing to produce, and how language and material are organized. What changes with generative systems is that part of production is taken directly into the system, and the scope of deliverable content and executable activity widens. How much existing material it inherits, and in what manner, still depends on the concrete processes of data acquisition and processing; the internet as a whole cannot simply be equated with some model's training set.

Within these combinations, how the window of verification changes depends on the concrete connection of design and institution. Completing an operation automatically can shorten waiting, and can also cost the user the opportunity to compare and to withdraw; confirmation steps, stable records, and arrangements for recourse can in turn change these consequences. Stopping, comparing, and correcting are not possessions that one era enjoys by nature and the next loses all at once; they are conditions of practice that need continuous maintenance.

§8.2 The Name Must Accommodate Counterexamples

If the concept of the Discrete Leviathan explains every successful service as the bait of dependence, every objection as proof that power has become invisible, and every reform as institutional self-protection, it becomes a name that cannot be corrected. The definition of this chapter must allow certain objects to be left out, and it must allow existing relations of control to weaken.

If an actor can migrate its activity at bearable cost, retaining the data, relationships, and business it needs, and the original node can no longer shape its conditions of participation, then the judgment of control ought to be narrowed accordingly. If an institution lets independent evidence actually change the allocation of resources and the rules, the relevant correction mechanism cannot be described as wholly ineffective. If differences in distribution do not impede common records and coordination, one cannot, from personalization alone, declare the public observation interface to have collapsed.

The analysis of time and belief must also accept more specific tests in reverse. If participants return to long-term work after resource guarantees are restored, that should raise the weight of the resource-constraint explanation; their earlier short-term choices alone do not establish a change in stable time preference. If different organizations of information did not change judgments about what others know, neither can one claim, from the mere difference of connection graphs, that common knowledge has been damaged. Such results would change the explanation of the mechanism, not merely the tone of the criticism.

These counterexamples demand no perfect real institution. What they test are concrete judgments of control and of loss, not an all-or-nothing line drawn at total liberation. Likewise, that a reform added some capability does not mean the remaining problems have disappeared. The analysis should be able to acknowledge local improvement and continuing dependence at once, stating the grounds for each.

Historical comparison is also subject to this discipline. Structural similarity can inspire questions; it cannot prescribe answers in advance. An earlier case may show how a relation was maintained; it may equally show the conditions under which it was changed.

§8.3 Discrete Control Facing Continuous Life

The Discrete Leviathan is a cross-layer network of linked organizations that continuously stipulates information, opportunity, and the paths of return, while placing dependent actors in relations of markedly asymmetric exit costs. Its power appears not only in the provision of services to existing preferences, but in its taking part in forming the conditions of preference and belief: what appears often, what can be made good, which consequences are seen, and how long a person still has to compare and to wait. It need not become a unified sovereign, nor need all participants believe it correct, in order to shape action through interconnected rules.

The historical materials show that such relational changes can run in different directions. Education gave knowledge a form of sustained transmission; habit spared belief from being rebuilt every time; intertemporal commitment enabled people to act for the future; the same organizational capacities could also let an established method shut out dissent, let familiarity stand in for grounds, and let one party's gains obscure another party's long-term life. What matters is not a longer or a shorter time horizon, but how the future is understood, who arranges it, and through what relations present commitments can be verified and borne.

The human beings this network faces are not a string of unrelated clicks. Completing work, developing professional competence, and sustaining relationships require ongoing effort, and responsibilities need someone to go on bearing them after the results appear. A business system can update account classifications, adjust resources, and replace services; a person may not be able to rebuild a life at the same speed. Short-term reactions can adapt to a set of rules, yet they cannot by themselves prove that a person's life has thereby improved.

The Clock Room of Chapter 1 here takes on a relational question: who arranges the clock face, who controls the path to external reference, and who has the right to change action after verification. The fiber analysis of Chapter 4 further shows that the same short-run choices are compatible with different preferences, beliefs, and circumstances; the accounting discipline of Chapter 3 then tracks how these relations accumulate costs across time, preventing immediate gains from settling the account on behalf of long-term consequences.

Maintaining the capacity for independent judgment therefore requires continuous work, and it requires relational arrangements in which that work has time, resources, rights, and consequences. The explanatory power of

the concept of the Discrete Leviathan depends on whether it can identify these concrete connections and their changes—not on whether it portrays every change in human life as following one irreversible direction of decay.

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